Battery-Powered DC–DC Buck–Boost Converter with Two Control Methods and a Single Dashboard Interface

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***Abstract*—Battery-powered electronic systems are prone to voltage fluctuations due to the natural discharge of batteries and variations in load. These fluctuations can cause the system to perform unpredictably or even fail if not properly regulated. To address this challenge, DC–DC converters are commonly used in these systems to stabilize and maintain a steady output voltage [1], [2]. This report explores the design and simulation of an inverting buck–boost converter, which has the unique ability to step-up or step-down the input voltage, making it ideal for battery-powered applications where the input voltage may vary [3].**

Two control methods were implemented and compared to manage the output voltage: a traditional proportional-integral (PI) controller, and a more advanced cascaded control scheme. In the cascaded method, a voltage PI controller governs the outer loop, while a current PI controller is used in the inner loop to provide more precise control [4]. The system was modeled and simulated using MATLAB/Simulink, which allowed for detailed analysis of the converter's performance under different operating conditions. A dashboard interface was also created for real-time monitoring and tuning, allowing for adjustments to improve system performance [7]. Simulation results demonstrated that the cascaded control method offers improved stability and quicker response times when compared to the simpler PI controller, especially under varying load and input conditions [6].

I. Introduction

Battery-powered electronic devices often experience fluctuations in input voltage due to factors such as battery discharge and load variations. These fluctuations can significantly impact system performance, causing instability or even system failure if voltage regulation is not properly implemented. Voltage regulation is crucial to ensure the reliable operation of these devices, especially in applications such as portable electronics and renewable energy systems. DC–DC converters are widely employed in power electronics to maintain a stable output voltage, compensating for input voltage variations [1], [2]. These converters offer efficient solutions for regulating the voltage, thus improving the performance and reliability of battery-powered systems.

Among the various topologies of DC-DC converters, the buck–boost converter is particularly advantageous because it can increase or decrease the input voltage depending on the operating conditions. This versatility makes the buck–boost converter ideal for applications where the input voltage can be higher or lower than the desired output voltage. Unlike buck or boost converters, which can only step down or step up the voltage, respectively, the buck–boost converter offers both capabilities, thus providing a more flexible solution for managing fluctuating supply voltages [3], [4]. This feature is particularly valuable in battery-powered systems where the supply voltage is subject to changes throughout the device's operation.

This report focuses on the design and simulation of an inverting buck–boost converter, exploring the implementation of two distinct control strategies to regulate the output voltage. The first approach uses a simple proportional-integral (PI) controller, which adjusts the duty cycle of the converter to maintain a steady output voltage. The second approach utilizes a cascaded voltage-current PI controller, which incorporates a voltage control loop in the outer loop and a current control loop in the inner loop to achieve faster and more precise control [5], [6]. The converter's performance is simulated using MATLAB/Simulink, and a dashboard interface is developed to allow for real-time monitoring and tuning of the system parameters [7]. This simulation enables a comprehensive analysis of both control methods, offering insights into their respective advantages and limitations under varying operating conditions.

II. Buck–Boost Converter Topology

The inverting buck–boost converter consists of the following components:

- Input voltage source (V_{in}): Provides the power to the converter.
- Inductor (L): Stores energy during the ON phase of the switch and releases it during the OFF phase.
- MOSFET switch (Q): The primary switching element that controls the flow of current.
- Diode (D): Ensures unidirectional current flow from the inductor to the load.
- Output capacitor (C): Filters the output to reduce voltage ripple.
- Load resistor (R): Represents the load connected to the converter.
- Electrical reference or Ground.

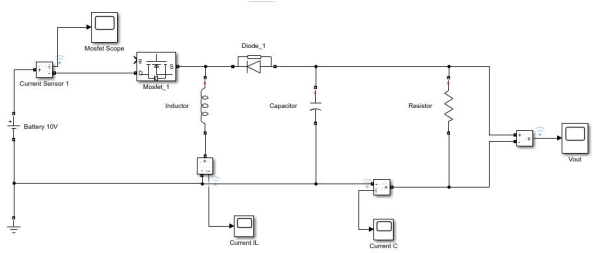

Fig. 1. Inverting buck–boost converter topology. This diagram illustrates the key components: MOSFET switch (Q), diode (D), inductor (L), capacitor (C), and load resistor (R). The negative polarity of the output is highlighted.

The voltage conversion ratio in continuous conduction mode (CCM) is:

$$V_o = -\frac{D}{1-D} V_{in} \tag{1}$$

where D is the duty cycle. The negative sign indicates an inverted output voltage.

A. Operating Principle

In a **buck-boost converter**, the inductor stores and releases energy during two switching phases.

B. Phase 1: Switch ON (Energy Storage)

When the **MOSFET switch turns ON**:

- The input voltage is directly connected to the inductor.
- Current starts **flowing through the inductor**.
- The diode becomes **reverse biased**.
- The load is temporarily powered by the **output capacitor**.

The current **increases linearly** because of the applied voltage:

$$\frac{di_L}{dt} = \frac{V_{in}}{L} \tag{2}$$

This increasing current builds a magnetic field, which means the inductor is storing energy.

So during this phase:

- Current increases
- Magnetic field grows
- Energy stored in the inductor increases

C. Phase 2: Switch OFF (Energy Released)

When the **MOSFET switch turns OFF**:

The current path through the switch is suddenly broken. An inductor does not allow instantaneous change in current. To keep current flowing, the inductor reverses its voltage polarity.

- The diode becomes forward biased.
- The inductor releases its stored energy to the output capacitor and load
- The current now decreases, meaning the magnetic field collapses, and stored energy is released.

$$\frac{di_L}{dt} = \frac{V_o}{L} \tag{3}$$

D. System Analysis

Transfer Function and Duty Cycle:

$$\frac{V_{out}}{V_{in}} = -\frac{D}{1-D} \tag{4}$$

$$D = \frac{V_{out}}{V_{out} - V_{in}} \tag{5}$$

Conservation of Power:

$$V_{in} \cdot I_{in} = V_{out} \cdot I_{out} \tag{6}$$

$$\begin{aligned} I_{in} &= \frac{-D}{1-D} \cdot I_{out} \\ &= \frac{-D}{1-D} \cdot \frac{V_{out}}{R} \\ &= (\frac{-D}{1-D})^2 \cdot \frac{V_{in}}{R} \end{aligned} \tag{7}$$

Inductor Current:

$$\begin{aligned} I_L &= \frac{I_{in}}{D} \\ &= \frac{D}{(1-D)^2} \cdot \frac{V_{in}}{R} \end{aligned} \tag{8}$$

III. Converter Design Methodology

A. Design Specifications

TABLE I
CONVERTER DESIGN SPECIFICATIONS

Parameter	Value
Input Voltage	10 V
Output Voltage	(-20) - (-5) V
Switching Frequency	20 kHz
Load Resistance	10-200 Ω

B. Duty Cycle Calculations

Using (5), the duty cycle can be calculated for edge cases V_{out}= -20V and V_{out} = -5V.

$$Dmax = \frac{-20}{-20-(-10)}$$
$$Dmax = 2/3$$

$$Dmin = \frac{-5}{-5-(-10)}$$
$$Dmin = 1/3$$

$$1/3 <= D <= 2/3 \tag{9}$$

C. Inductor Selection

Inductance is chosen to maintain continuous current mode (CCM) in all output voltages and resistance loads. Circuit analysis of the switch ON state yields the following inductor formula:

$$L = \frac{V_{in} D}{\Delta I_L f_s} \tag{10}$$

where ΔI_L is the peak to peak current ripple.
To ensure that CCM is maintained across all design parameters, the lowest possible inductor current must be considered. (8) shows I_L is a function of Duty Cycle and resistance.

$$\begin{aligned} I_{Lmin} &= I_L(D_{min}, R_{max}) \\ &= \frac{1/3}{(2/3)^2} \cdot \frac{10}{200} \\ I_{Lmin} &= 37.5mA \end{aligned} \tag{11}$$

$$\begin{aligned} \Delta I_L &= 2 \cdot I_{Lmin} \\ \Delta I_L &= 75mA \end{aligned} \tag{12}$$

Using (10), the Inductance is obtained:

$$\begin{aligned} L &= L(D_{min}, \Delta I_L) \\ L &= 2.22mH \end{aligned} \tag{13}$$

D. Capacitor Selection

Capacitance is chosen to ensure a maximum voltage ripple of 5% across all design parameters. Circuit analysis of the switch ON state yields the following formula:

$$C = \frac{I_{out} D}{\Delta V f_s} \tag{14}$$

setting ΔV to 0.05 V_{out} yields:

$$\begin{aligned} C &= \frac{I_{out} D}{0.05 V_{out} f_s} \\ &= \frac{D}{0.05 R f_s} \end{aligned} \tag{15}$$

Using (15), the Capacitance is obtained:

$$\begin{aligned} C &= C(D_{max}, R_{min}) \\ &= \frac{2/3}{0.05 \cdot 10 \cdot 20000} \\ C &= 66.7uF \end{aligned} \tag{16}$$

IV. Controller Design and Tuning

A. PI Voltage Controller

The PI controller regulates the output voltage:

$$u(t) = K_p e(t) + K_i \int e(t) dt$$

where $e(t) = V_{ref} - V_o$.

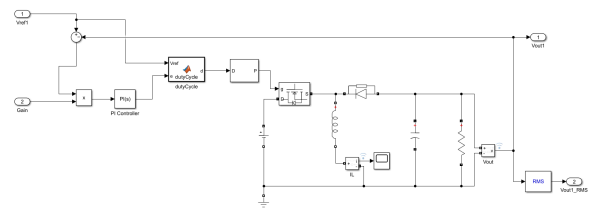

Fig. 2. Voltage PI Controller diagram. Shows feedback from output voltage to adjust duty cycle for stability.

Tuning via the Ziegler-Nichols Method:

1) Set $K_i = 0$ for open-loop P-control.
2) Increase K_p until continuous oscillations appear.
3) Measure the ultimate gain K_u and period P_u.
4) Controller gains are then calculated for a PI controller:

$$K_p = 0.45K_u, \quad K_i = \frac{0.54K_p}{P_u}$$

Bench simulations yield:

$$K_u = \frac{5}{3} \tag{17}$$

$$P_u = 4.84ms \tag{18}$$

$$K_p = 0.75 \tag{19}$$

$$K_i = 186 \tag{20}$$

B. Cascaded Voltage–Current PI Controller

The cascaded controller includes a fast inner current loop and an outer voltage loop. The outer loop provides the expected inductor current based on voltage error, while the inner loop adjusts the duty cycle based on inductor current error.

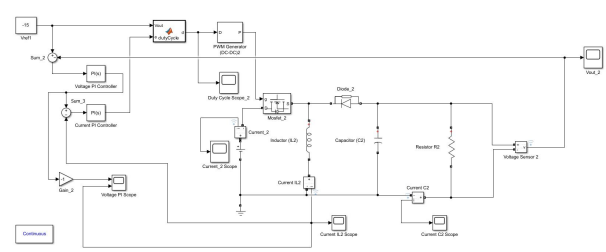

Fig. 3. Cascaded Voltage-Current PI Controller.

Cascaded PI tuning: Since the inner-loop adjusts the error based on current readings through the inductor, this allows the inductor to be treated as a basic integrator with plant:

$$G_i(s) = \frac{1}{Ls} \tag{21}$$

Using the direct gain formula:

$$w_c = \frac{w_s}{10} = 2 \cdot \pi \cdot (2000) \tag{22}$$

$$K_p = Lw_c = 27.632 \tag{23}$$

$$K_i = \frac{K_p w_c}{4} = 86764.48 \tag{24}$$

Eliminating the inductor from the outer-loop makes it a simple integrator with plant:

$$G_v(s) = \frac{1}{Cs} \tag{25}$$

Using the direct gain formula:

$$w_v = \frac{w_c}{10} = 2 \cdot \pi \cdot (200) \tag{26}$$

$$K_{pv} = Cw_v = 0.0837 \tag{27}$$

$$K_{iv} = \frac{K_v w_v}{4} = 26.305 \tag{28}$$

C. Duty Saturation

In the duty cycle function, the duty cycle is saturated:

$$0.25 <= D <= 0.75$$

This allows the controller approximately 20% of error adjustment above minimum/maximum duty cycle values. This limitation is important as high duty cycles can cause major voltage and current spikes, while low duty cycles will impact power conservation.

V. System Simulations

A. Simulation Setup

The system was modeled and simulated using MATLAB Simulink, which allowed for in-depth analysis of inductor current, output voltage and system behavior across all design parameters.

The system was tested with no control, PI control and the Cascaded control. For each setup, the test was run with loads R_{min}=10 Ω and R_{max}= 200 Ω. Additionally, each test involved step-up and step-down transitions to observe the transient response, overshoot and settling time.

B. Simulation Results

Worst case voltage ripple occurs at D_{max}, R_{min}, this was confirmed to be with-in design parameters ($V_{ripple} < 0.05V_{out}$) for both control methods (Figure 4 & Figure 5).

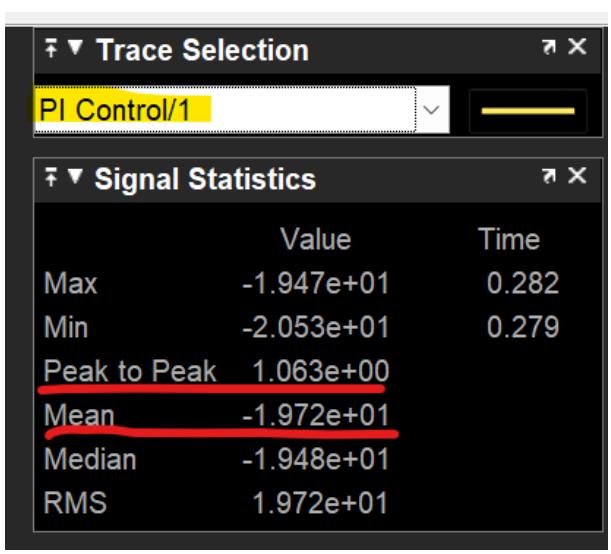

Fig. 4. Voltage Ripple at PI Control

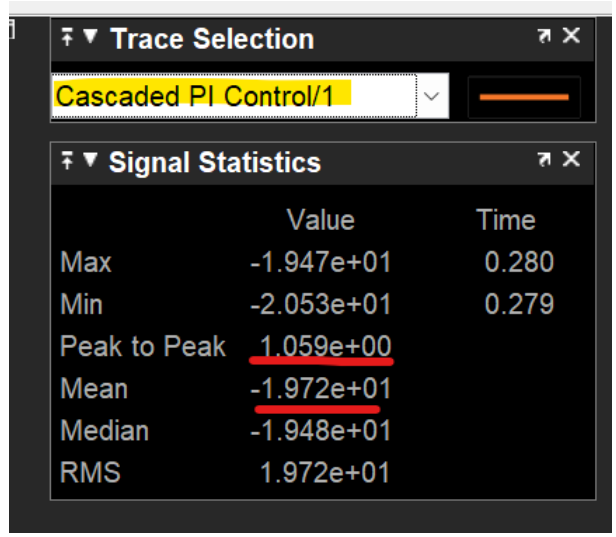

Fig. 5. Voltage Ripple at Cascaded PI Control

CCM was verified for worst current conditions D_{min}, R_{max}) for both control methods. (Figure 6 & Figure 7).

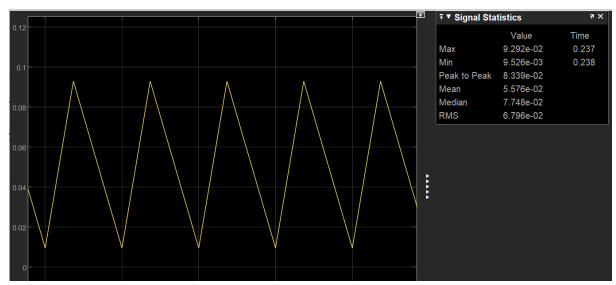

Fig. 6. PI: CCM maintained at lowest possible inductor current

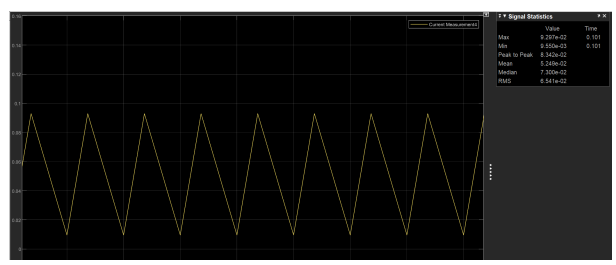

Fig. 7. Cascaded: CCM maintained at lowest possible inductor current

During the simulation, the key performance metrics output voltage ripple, overshoot, and settling time were recorded for both controllers under different load conditions (10Ω and 200Ω).

Without any added control, steady state error was observed at low and high loads (Figure 8 & Figure 9). At higher loads, output voltage ripple is reduced, but major step down overshoot was observed. This major overshoot could damage the load. This is an acceptable starting point which maintains

an acceptable ripple and CCM across all parameters, steady state and overshoot must be addressed.

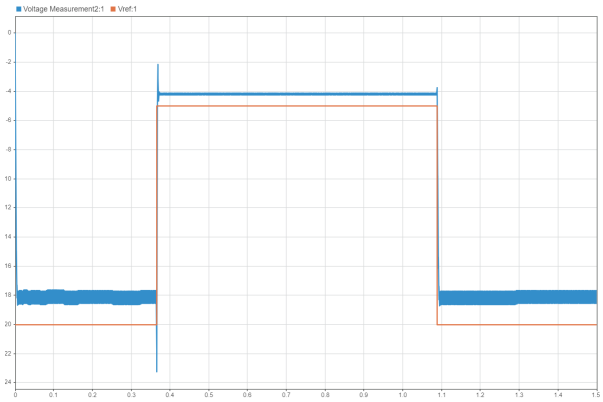

Fig. 8. No Control applied @ 10 Ω. Overshoot and steady-state error

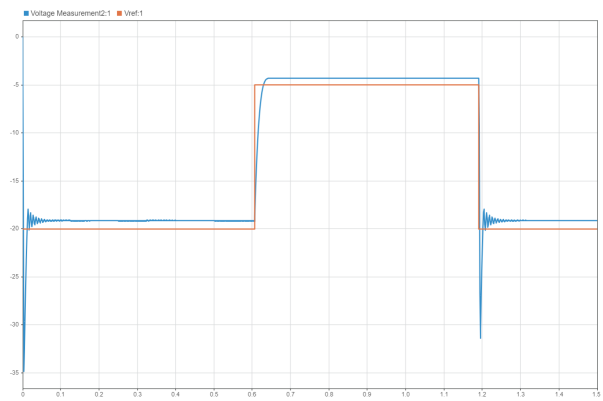

Fig. 9. No Control applied @ 200 Ω. Overshoot and steady-state error

The PI control resolved the steady state error, while maintaining the acceptable voltage ripple and CCM across all loads. However, at higher loads, system instability was observed, which needed correction via error gain adjustment. A gain slider is added to reduce the PI correction while maintaining the K ratios.

At higher loads, with adjusted gain, the PI controller experiences significantly less voltage ripple, with large overshoots and longer settling times.

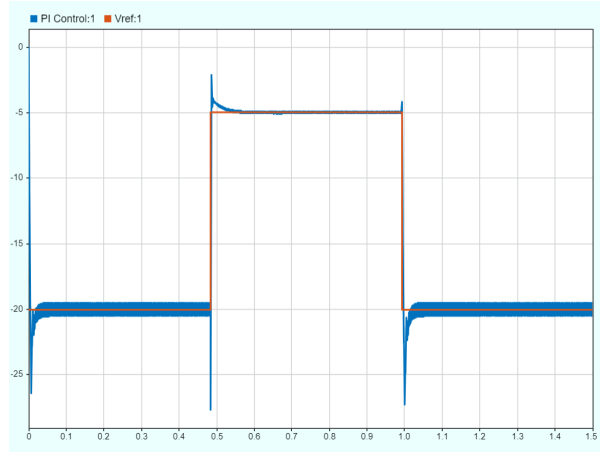

Fig. 10. PI control @ 10 Ω. Steady-state error minimized.

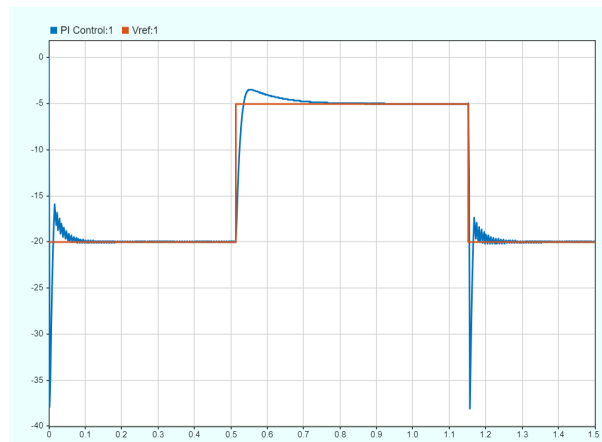

Fig. 11. PI controller @ 200 Ω. Adjusted gain, minimizes steady-state error but increases overshoot and settling time.

At low loads, the Cascaded PI controller resolved the steady state error, while introducing smoother transitions with less overshoot. However, it really shines with an adjusted gain at high loads, producing almost no voltage ripple, and very minimal overshoots and transition responses.

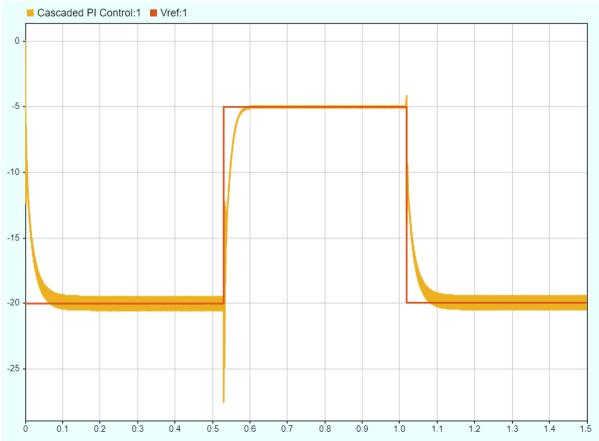

Fig. 12. Cascaded PI control @ 10 Ω. Smoother transitions with reduced overshoot.

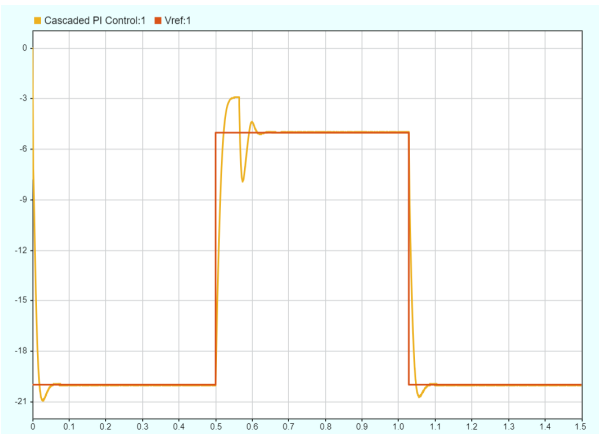

Fig. 13. Cascaded PI controller @ 200 Ω. Significantly reduced step-down overshoots with almost no voltage ripple.

Comparing both control systems at low loads shows a comparable result. The Cascaded controller has smoother but slower transitions. At high loads, the cascaded controller shines and completely outperforms. It has significantly faster settling time, almost no overshoot, and no voltage ripple.

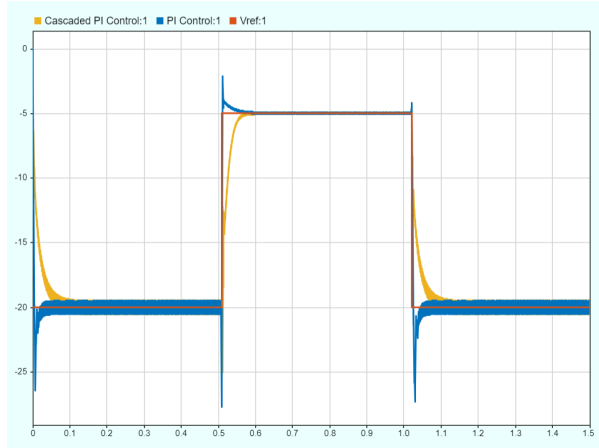

Fig. 14. Comparison PI vs Cascaded @ 10 Ω. Cascaded controller provides faster settling and lower overshoot.

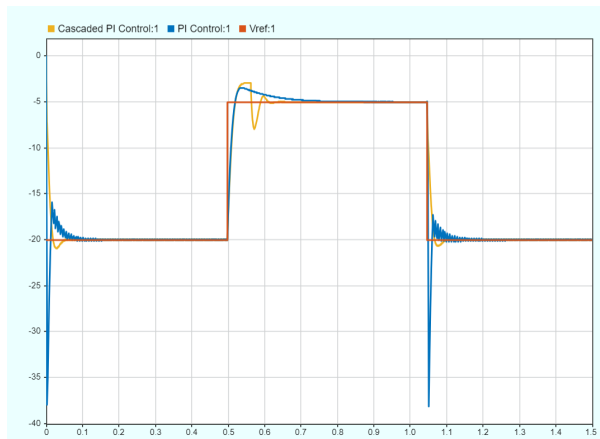

Fig. 15. Comparison PI vs Cascaded @ 10 Ω. Cascaded controller provides faster settling and lower overshoot.

VI. Discussion

TABLE II
Controller Performance Comparison

Metric	Cascaded Controller	PI Controller
Steady-State Error	Very Low	Low
Voltage Ripple	Low	Moderate
Overshoot	Small	Moderate
Settling Time	Fast	Slightly Slower

A. Load Variations

To understand how the system performs under different conditions, the load resistance R was varied between 10Ω and 200Ω. These variations simulate both light and heavy load scenarios, helping to understand how the converter adjusts to changes in power demand. In a light load condition** (with a high resistance of 200Ω), the system only needs to provide a small current to the load. This results in a lower current through the inductor, which may cause higher voltage ripple due to reduced current flow [5]. On the other hand, in a heavy

load condition (with a low resistance of 10Ω), the converter must supply a much higher current, stressing the system more. Under this condition, the control system needs to adjust more frequently to maintain the output voltage and avoid excessive voltage fluctuations.

By varying the load, we can simulate real-world conditions where a device's power demand isn't constant. For example, in battery-powered applications such as smartphones or laptops, the load changes depending on the tasks being performed. The converter must adapt to these fluctuations while maintaining stable voltage to ensure reliable performance. This aspect of the simulation is crucial because it provides insight into how well the converter handles changes in power demand, which can affect both the output voltage regulation and the overall efficiency of the system [6].

B. Performance Evaluation

To assess the converter's performance, several key factors were considered:

- **Voltage Regulation**: The primary goal of any DC–DC converter is to maintain a stable output voltage despite fluctuations in input voltage or load resistance. In both light and heavy load conditions, the system's ability to regulate the output voltage and minimize voltage ripple was closely monitored. The converter performed well under varying load conditions, with the voltage staying within acceptable limits, even when the load changed dramatically [7].
- **Ripple and Noise**: Voltage ripple refers to small fluctuations in the output voltage caused by the switching process. These fluctuations are more noticeable under lighter loads, where the converter has to work harder to maintain a stable voltage. Excessive ripple can lead to issues in sensitive electronic systems, so minimizing it was a key design consideration. The simulation results showed that the ripple was kept within acceptable limits, especially under higher load conditions, where the higher current helped smooth out the fluctuations [8].
- **Efficiency**: The efficiency of the converter was evaluated by comparing the input power and output power under different load conditions. The converter was designed to be highly efficient, but efficiency can drop under heavier loads due to higher conduction losses in the inductor and switch. The simulations showed a slight decrease in efficiency at low loads, which is expected due to lower current levels and less optimal operation of the converter's components [9].
- **Transient Response and Settling Time**: One of the most important aspects of the converter's performance is how quickly it can respond to changes in load or input voltage. Settling time refers to how quickly the output voltage stabilizes after a disturbance, while the transient response reflects how well the converter can adjust to the change. The cascaded control scheme used in this study showed a faster response time and smaller overshoot compared to the simple PI controller, thanks to its dual-loop control structure. This approach allowed the converter to quickly adapt to load variations and minimize both overshoot and settling time [10], [11].

VII. Conclusion

The inverting buck–boost converter was successfully designed and simulated using MATLAB/Simulink. The cascaded controller demonstrated superior voltage regulation, lower ripple, and faster transient response compared to a single PI controller. Proper tuning using both Ziegler-Nichols and Frequency Domain methods significantly improved system performance.

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